

Government Engineering College Kishanganj

Department of Computer Science and Engineering

Computer Networks LAB-03

Objective

Develop a TCP-based client-server chat application using socket programming in C.

Problem Statement

Write a C program to implement a **client-server chat application** using TCP sockets with the following requirements:

- The server must listen on **port 8081**.
- The client connects to the server using its IP address.
- Both client and server should be able to send and receive messages.
- Communication continues until the client sends the message **"bye"**.
- Upon receiving **"bye"**, the server should terminate the connection gracefully.

Additional Requirements

- Use TCP sockets (`SOCK_STREAM`).
- You may use the `string.h` library.
- Execute the program:
 - On the same machine (localhost / 127.0.0.1)
 - On different machines (using actual IP address)

Server Template

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 8081
#define BUFFER_SIZE 1024

int main() {

    int server_fd, new_socket;
    struct sockaddr_in address;
    int addrlen = sizeof(address);

    char buffer[BUFFER_SIZE];
    char message[BUFFER_SIZE];

    // Create socket
    server_fd = socket(AF_INET, SOCK_STREAM, 0);

    if (server_fd < 0) {
        perror("Socket creation failed");
        exit(1);
    }

    // Configure address
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);

    // Bind socket
    if (bind(server_fd, (struct sockaddr *)&address,
        sizeof(address)) < 0) {
        perror("Bind failed");
        exit(1);
    }
}
```

```

// Listen for connections
if (listen(server_fd, 5) < 0) {
    perror("Listen failed");
    exit(1);
}

printf("Server is listening on port %d...\n", PORT);

// Accept connection
new_socket = accept(server_fd, (struct sockaddr *)&
    address, (socklen_t *)&addrlen);

if (new_socket < 0) {
    perror("Accept failed");
    exit(1);
}

printf("Client connected.\n");

// Communication loop
-----
-----
-----

close(new_socket);
close(server_fd);

return 0;
}

```

Client Template

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

```

```

#define PORT 8081
#define BUFFER_SIZE 1024

int main() {

    int sock;
    struct sockaddr_in serv_addr;

    char buffer[BUFFER_SIZE];
    char message[BUFFER_SIZE];

    // Create socket
    sock = socket(AF_INET, SOCK_STREAM, 0);

    if (sock < 0) {
        perror("Socket creation failed");
        exit(1);
    }

    // Configure server address
    serv_addr.sin_family = AF_INET;
    serv_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, "127.0.0.1", &serv_addr.sin_addr);

    // Connect to server
    if (connect(sock, (struct sockaddr *)&serv_addr,
        sizeof(serv_addr)) < 0) {
        perror("Connection failed");
        exit(1);
    }

    printf("Connected to server.\n");

    ----
    ----
    ----

    close(sock);
    return 0;
}

```